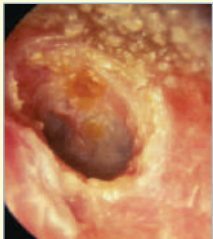


EAR/HEARING SYMPTOMS

Ask about ear itch, discharge from ear, ear pain or difficulty hearing/tinnitus (ringing/buzzing in ear/s). Then look in ear.

Itchy ear

Redness, swelling and/or pus in ear canal



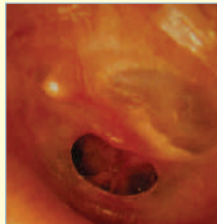
©University of Cape Town

Otitis externa likely

- Clean ear.¹
- After cleaning, instil **acetic acid 2% in alcohol** 4 drops in ear 6 hourly for 5 days.
- Give **paracetamol** 1g 4-6 hourly (up to 4g in 24 hours) as needed for up to 5 days.
- If severe pain, firm red swelling or temperature $\geq 38^{\circ}\text{C}$, give **flucloxacillin**² 500mg or **cefalexin** 500mg 6 hourly for 5 days.
- Refer if:
 - No better after 5 days
 - Blisters on ear, **herpes zoster** likely
 - Red swollen painful ear lobe, **cellulitis** likely

Discharge from ear

Symptoms ≥ 2 weeks, hole in eardrum



©University of Cape Town

Chronic suppurative otitis media likely

- Clean ear¹ repeatedly.
- If poor response to treatment, test for HIV $\hookrightarrow$ 110 and TB $\hookrightarrow$ 92.
- Refer if:
 - No better after 4 weeks
 - Hole in eardrum large, not getting smaller after 3 months, or persists > 6 months.
 - Difficulty hearing
 - Yellow/white deposit on eardrum, **cholesteatoma** likely.
- Refer same day if:
 - Painful swelling behind ear, **mastoiditis** likely
 - Neck stiffness

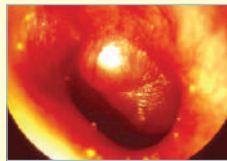
Symptoms for < 2 weeks

Painful ear

- If ear also itchy, consider otitis externa (see adjacent).
- Able to view eardrum?

Yes

- If normal looking ear, **referred pain** likely, check mouth and face:
 - If gum or tooth problem $\rightarrow$ 36.
 - If painful swelling of one/both sides of face, **mumps** likely $\rightarrow$ 32.
 - If pain in temporomandibular joint, check for joint problem $\rightarrow$ 62.
- If red bulging eardrum, **acute otitis media** likely:



©University of Cape Town

Acute otitis media likely

- Give **paracetamol** 1g 4-6 hourly (up to 4g in 24 hours) as needed for up to 5 days.
- Give **amoxicillin**² 1.5g 12 hourly for 5 days. If patient has had amoxicillin in last 30 days: give instead **amoxicillin/clavulanic acid**² 875/125mg 12 hourly for 5 days.
- If nose symptoms, consider and treat for **allergic rhinitis** $\hookrightarrow$ 34.
- If discharge, clean ear¹ and avoid getting ear wet.
- If recurrent episodes, test for HIV $\hookrightarrow$ 110 and refer.
- If no response to treatment after 3 days, refer.
- Refer same day if: neck stiffness or painful swelling behind ear, **mastoiditis** likely

No

Any of:

- Pain > 2 days
- Pain that wakes patient at night
- Temperature $\geq 38^{\circ}\text{C}$ in past 2 days

Yes

Treat for **acute otitis media**:

No

- Give **paracetamol** 1g 4-6 hourly (up to 4g in 24 hours) as needed for up to 5 days.
- If no better in 2 days, advise to return: treat for **acute otitis media**:

Difficulty hearing or tinnitus

- If on amikacin, discuss with TB doctor.
- If itchy/painful ear or discharge from ear, see adjacent column/s.
- Look in ear for foreign body and wax:

Foreign body

Wax

- Syringe ear/s with warm water.
- Avoid syringing and refer instead if:
 - Hole in eardrum
 - Chronic suppurative otitis media
- If unsuccessful after 3 attempts or causes pain, stop and refer/discuss with doctor.
- If hearing no better after foreign body/wax removal, refer for hearing test.

Normal looking ear

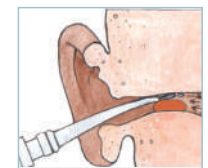


©University of Cape Town

- Arrange hearing test.
- Look for cause: Ask about prolonged exposure to loud noise.
- Review medication: aspirin, NSAIDs and furosemide.
- Refer if:
 - Sudden onset
 - One-sided
 - Dizziness/vertigo
 - Patient taking amikacin

How to syringe an ear

Fill a large syringe (50-200mL) with warm water. Ask patient to hold container under ear against neck to catch water. Gently pull ear upwards and backwards to straighten ear canal. Place tip of syringe at ear canal opening (no further than 8mm into canal) and direct water spray upwards in ear canal.



¹Cleaning the ear (dry mopping): roll a piece of clean soft tissue into a wick. Carefully insert wick into ear with twisting action. Remove wick and replace with clean dry wick. Repeat until wick is dry when removed. Never leave wick or other object inside ear. The ear can only heal if dry. ²If severe penicillin allergy (history of anaphylaxis, urticaria or angioedema), give instead **azithromycin** 500mg daily for 3 days.